

# SPENCER MICHAELSON

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Portfolio: <https://smichaelson.github.io/>

## EDUCATION

**Harvey Mudd College**, Claremont, CA  
B.S. in Engineering | GPA: 3.912/4.000

May 2028

## PROFESSIONAL EXPERIENCE

**Engineering Intern III**, Lean Technology Corporation, Costa Mesa, CA

May 2026 - August 2026

- *Promoted from Engineering Intern I (May 2025 - August 2025) and Engineering Intern II (August 2025 - May 2026).*
- Modeled **electrical fault scenarios** and **circuit-network behavior** for major airport infrastructure programs at SFO and SEA-TAC.
- Built a QGIS workflow integrating 360° field video, survey data, and geospatial layers to accelerate field verification, spatial analysis, and design coordination for airport electrification projects.
- Created Python/SQL project-management and electrical forecasting tools for a **\$1.6B SFO electrification and expansion portfolio**, supporting planning across 250+ infrastructure projects under V&V-driven workflows.
- Automated airport traffic modeling, regulatory/weather/RF/geologic data analysis, and FAA documentation workflows using Python and SQL, **reducing turnaround time by 5x**.

## TECHNICAL LEADERSHIP

**Chief Engineer**, Mudd Amateur Rocketry Club (MARC), Harvey Mudd College

August 2024 - Present

- *Promoted from Structures Member (August 2024 - August 2025) and Propulsion Lead (August 2025 - March 2026).*
- Served as Chief Engineer for **Apollyon I**, a 6-inch, 60-lb, full-M-class **FAR Unlimited 2026** vehicle; led a 40-member team across propulsion, structures, avionics, simulation, recovery, payloads, integration, and launch operations, earning **1st place**.
- Owned vehicle architecture, requirements flowdown, integration schedule, technical reviews, and launch-readiness planning across propulsion, structures, avionics, recovery, payload, and flight simulation.
- Directed trade studies for motor selection, stability margin, max-Q, recovery sizing, and composite airframe design leveraging OpenRocket, RASAero II, MATLAB, SolidWorks, and COMSOL; downselected an architecture based on stability, recovery mass, and competition constraints.
- Fabricated and structurally integrated **Gladius III**, contributing to **3rd place at FAR Unlimited 2025**; validated flight simulation against altimeter data with **99.95% apogee prediction accuracy**.

**Chief Technical Analyst**, Claremont Aerospace & Defense Association (CADA), Harvey Mudd College

December 2025 - Present

- Led aerospace/defense technical analyses, translating stakeholder objectives into payload concepts, mission architectures, CONOPS, integration constraints, and risk assessments for autonomous system applications.
- Directed analysis of current arms manufacturing processes and developed an additional product stack while accounting for regulations, existing facilities, and competition.

**President**, Mudd Automotive Club Hybrids (MACH), Harvey Mudd College

August 2024 - Present

- *Promoted from Mechanical Member (August 2024 - August 2025) and Safety Officer (August 2025 - April 2026).*
- Lead a 13-member **hybrid-electric vehicle team** across mechanical design, electrical integration, manufacturing, safety, and testing workstreams.
- Oversee design reviews, the build schedule, and subsystem integration for chassis, braking, steering, and hybrid-powertrain systems.
- Designed and manufactured frame, braking, and steering assemblies in SolidWorks using DFM principles and FEA to evaluate safety factor, stiffness, and load paths.

## SELECTED PROJECTS

Independent Project, **Hat Trick**, Harvey Mudd College

February 2025 - March 2026

- Designed, built, and flew **Hat Trick** through **6** high-power launch campaigns to 7,000 ft on K-class motors, validating recovery, avionics, and structural integration across **iterative flight tests**.
- Developed a **dual-redundant avionics stack** for flight data logging, deployment control, and recovery reliability using multi-sensor fusion, LoRa telemetry, and custom enclosures.
- Implemented onboard **GPS tracking and a post-flight data pipeline** to reconstruct trajectory, evaluate recovery performance, and compare flight data against simulation predictions.
- Analyzed recovered flight data and post-flight inspection results to improve avionics packaging, recovery reliability, and launch integration procedures.

Independent Project, **Unknown Rider**, Harvey Mudd College

March 2026 - Present

- Developing **Unknown Rider**, a NAR Level 3 **certification vehicle and avionics test platform** for high-altitude flight-data collection, recovery validation, and embedded-system testing.
- Performing preliminary aerodynamic, structural, and RF link-budget analyses to assess stability margin, airframe survivability, telemetry range, and recovery architecture for projected **50,000+ ft / Mach 2+** flight conditions.

Independent Project, **Atlas Flight Guidance Computer**, Harvey Mudd College

March 2026 - Present

- Designing **Atlas**, a custom STM32-based flight-guidance computer for state estimation, active control, data logging, recovery deployment, live telemetry, GPS tracking, peripheral expansion, and guidance-interface research for high-power rocketry.
- Manufactured the system and wrote system firmware using FreeRTOS, confirming **full functionality** of actuator control, sensors, Bluetooth, LoRa, GNSS, USB-C, buzzer, local storage, and GPIO under load.

## RESEARCH EXPERIENCE

**Undergraduate Researcher**, Harvey Mudd College: Drone Lab, Dr. Jason Gallichio

January 2025 - December 2025

- Created **drone guidance and flight-analysis tools** for radio telescope beam mapping, supporting autonomous flight planning, beam-prediction workflows, and post-flight trajectory analysis.
- Built a hardware-agnostic forward simulation to predict beam-mapping coverage and flight-path performance before field testing.
- Planned, executed, and analyzed **40+ autonomous flights**, tuning GNC behavior and evaluating trajectory repeatability using flight logs and post-processing scripts.

## SKILLS

**Programming:** MATLAB (10 years), Python (7 years), R (6 years), C/C++ (5 years), SQL (3 years) | **Development tools:** Git (5 years)

**Software:** SolidWorks, Simulink, COMSOL, Bluebeam, AutoCAD, pgAdmin, QGIS, KiCad, STM32 Ecosystem, OpenRocket, RASAero II

**Analysis:** Electrical fault simulation, FEA, CFD, flight simulation, post-flight analysis, GNC, trajectory reconstruction

**Manufacturing:** Additive manufacturing, manual machining, CNC machining, laser cutting, water jet, composites, GD&T

**Certifications:** NAR Level 1 & NAR Level 2 High-Power Rocketry